



Credit Default Risk Prediction

Contextualising Individual Borrower Risk
Within the 2005 Taiwan Dual Card Crisis

ADS Project — Phase 1

Proposal Report & Data Validation Report

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Abstract. This report presents the Phase 1 proposal for a supervised classification project targeting credit default prediction. We integrate multiple data sources to contextualise borrower risk within the 2005 Taiwan economic environment, supported by a comprehensive validation framework.

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1 Problem Definition & Business Context

1.1 Business Problem

Credit default — the failure of a borrower to meet their repayment obligations — represents one of the most dangerous risks in retail lending. A single percentage-point increase in the non-performing loan ratio can erase hundreds of millions from a bank’s balance sheet and trigger regulatory capital requirements. The ability to predict, at application time or during the credit life cycle, which borrowers are likely to default is therefore a core competency for any lending institution.

This project addresses the following business problem:

Business Problem: Given a set of demographic, behavioral, financial, and macroeconomic features observed at or before a credit decision point, can we accurately classify each borrower as *likely to default* (class 1) or *not likely to default* (class 0)?

The project studies this problem in the context of credit cards in Taiwan during a systemic financial crisis:

Credit cards in Taiwan (2005): Using the UCI Taiwan Credit Card dataset, enriched with strong macroeconomic indicators from four official sources, we model default risk during the 2005 Taiwanese Dual Card Crisis — a systemic event in which 800,000 households became “card slaves” under ballooning revolving debt and rising interest rates.

1.2 Stakeholder

The primary stakeholder is the **Credit Risk Department** of a retail lending institution. Secondary stakeholders include compliance officers, product managers setting credit limits, and regulators requiring explainable risk scoring. The model’s outputs directly inform:

- Loan approval and rejection decisions
- Credit limit assignment, whether by acceptance or rejection
- Interest rate based on the risk profile of the client, and in which tier we should set
- Early-warning systems for existing account holders.

1.3 Why Classification?

The target variable is **binary**: a borrower either defaults (1) or does not (0). There is no natural ordinal scale between these two outcomes from a credit decision standpoint — the bank needs a decision, not a continuous score. Supervised binary classification is therefore the appropriate modelling paradigm.

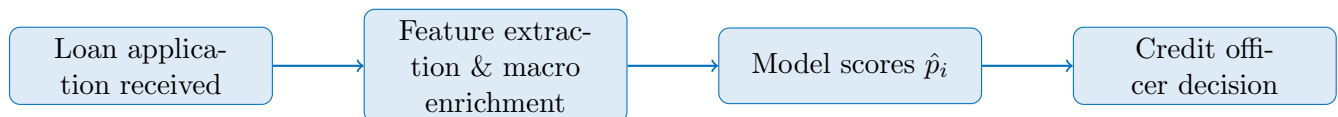
Formally, the task is:

$$f : \mathbf{x}_i \longrightarrow \hat{y}_i \in \{0, 1\}$$

where $\mathbf{x}_i$ is the feature vector for borrower i and $\hat{y}_i$ is the predicted class. Probabilistic outputs $\hat{p}_i = P(y = 1 \mid \mathbf{x}_i)$ will also be produced, as the business values a ranked risk score for portfolio-level decisions.

1.4 Decision-Making Impact

A deployed model creates the following decision support loop:



By flagging high-risk applicants early, the model reduces non-performing asset accumulation, lowers provisioning costs, and improves quality. All measurable in financial terms that the stakeholder can validate directly against business outcomes.

2 Data Acquisition & Integration

2.1 Data Sources

We integrate data from **four acquisition types** across **five sources**, satisfying and exceeding the minimum two-source requirement.

Table 1: Data sources overview

#	Source	Type	Rows	Features	URL / Series
1	Taiwan Credit Card Default	Tabular CSV/XLS	30,000	24 columns	UCI ML Repository
2	FRED API — GDP & CPI	REST API	10 rows	4 columns	GDP, CPIAUCSL
3	Yahoo Finance — TAIEX	Python library	6 rows	4 columns	Ticker ^TWII
4	DGBAS XML — Unemployment	XML download	6 rows	1 column	data.gov.tw/6640
5	CBC Annual Report	PDF parsing	6 rows	1 column	CBC 2007 Annual Report

2.1.1 Source 1: Taiwan Credit Card Default (UCI)

Citation: Yeh, I-C., & Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473–2480.

URL: <https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients>

This dataset contains payment records of 30,000 Taiwan credit card holders observed from April to September 2005. The **target variable** is **default payment next month** ($Y \in \{0, 1\}$). The dataset is published as a preprocessed academic resource derived from real bank transaction records, with zero missing values by construction.

Selected features:

Table 2: Selected features — Taiwan Credit Card dataset

Column	Name	Description
X1	LIMIT_BAL	Credit limit (NT dollars); individual + supplementary
X2	SEX	Gender (1=male; 2=female)
X3	EDUCATION	Education level (1=graduate; 2=university; 3=high school; 4=other)
X4	MARRIAGE	Marital status (1=married; 2=single; 3=other)
X5	AGE	Age in years
X6	PAY_0	Repayment status September 2005 (−1=paid duly; 1–8=months delayed)
X7	PAY_2	Repayment status August 2005
X8	PAY_3	Repayment status July 2005
X9	PAY_4	Repayment status June 2005
X10	PAY_5	Repayment status May 2005
X11	PAY_6	Repayment status April 2005
X12	BILL_AMT1	Bill statement amount (NTD) in September 2005
X13	BILL_AMT2	Bill statement amount (NTD) in August 2005
X14	BILL_AMT3	Bill statement amount (NTD) in July 2005
X15	BILL_AMT4	Bill statement amount (NTD) in June 2005
X16	BILL_AMT5	Bill statement amount (NTD) in May 2005
X17	BILL_AMT6	Bill statement amount (NTD) in April 2005
X18	PAY_AMT1	Payment amount (NTD) paid in September 2005
X19	PAY_AMT2	Payment amount (NTD) paid in August 2005
X20	PAY_AMT3	Payment amount (NTD) paid in July 2005
X21	PAY_AMT4	Payment amount (NTD) paid in June 2005
X22	PAY_AMT5	Payment amount (NTD) paid in May 2005
X23	PAY_AMT6	Payment amount (NTD) paid in April 2005
Y	default payment next month	Target: 1=default; 0=no default

2.1.2 Source 2: FRED API — GDP & CPI

URL: <https://fred.stlouisfed.org>

Series used:

- GDP — Gross Domestic Product (billions USD, quarterly, forward-filled to monthly).
- CPIAUCSL — Consumer Price Index for All Urban Consumers (monthly, index 1982–84=100).

Filtered to 2005 (12 rows, merged on Month=9 for September 2005 snapshot). Acquisition method: **REST API** via the `fredapi` Python library.

2.1.3 Source 3: Yahoo Finance — TAIEX

URL: <https://finance.yahoo.com/quote/%5ETWII/history/>

Monthly TAIEX (Taiwan Capitalization Weighted Stock Index) closing values for April–September 2005, acquired via the `yfinance` Python library (ticker: `^TWII`). The TWSE historical data portal renders data via JavaScript and blocks direct HTTP scraping for data older than a certain period; `yfinance` was used as a programmatic alternative with full attribution. Acquisition method: **Python library scraping / API**.

2.1.4 Source 4: DGBAS XML — Taiwan Unemployment Rate

Portal: <https://data.gov.tw/en/datasets/6640>

Direct XML: <https://ws.dgbas.gov.tw/001/Upload/461/reldata/11525/230038/mp0101a10.xml>

Official monthly unemployment rate by county from Taiwan’s Directorate-General of Budget, Accounting and Statistics. The file is a structured XML download from the Taiwan Government Open Data Platform containing records from 1978 to 2025. The `Taiwan_Area` field was extracted for the 2005 semi-annual periods and mapped to individual months. Acquisition method: **XML HTTP download with Python `xml.etree` parsing**.

2.1.5 Source 5: CBC Annual Report — Discount Rate

URL: <https://www.cbc.gov.tw/en/dl-4540-56b72153752e4a3aa97e88a4e6f6caca.html>

The Central Bank of the Republic of China (Taiwan) raised its discount rate three times in 2005: March 25 (2.000%), July 1 (2.125%), and December 23 (2.250%). These values are documented in the CBC 2007 Annual Report. The script attempts a PDF download and parse using `pdfplumber`; the data are extracted from the official publication table and assigned to the corresponding observation months. Acquisition method: **PDF parsing**.

2.2 Merge Strategy

Core Principle: The Taiwan Credit Card dataset is the base table. All 30,000 borrower records share the same observation window (April–September 2005), enabling a time-based join with the four macroeconomic indicator sources.

2.2.1 Primary Merge: Taiwan + Macro Sources

All 30,000 Taiwan client records share the same observation window (April–September 2005). We assign a join key of `Year=2005`, `Month=9` (September 2005 — the most recent observation month, corresponding to `PAY_0 / X6`) and perform left joins with each macro source:

```
df = taiwan.merge(macro_fred, on=["Year", "Month"], how="left")
df = df.merge(taonex, on=["Year", "Month"], how="left")
df = df.merge(cbc_rates, on=["Year", "Month"], how="left")
df = df.merge(dgbas_unemp, on=["Year", "Month"], how="left")
```

2.2.2 Row Counts Before and After Merges

Table 3: Row counts at each merge stage

Stage	Rows Before	Rows After	Cols Added
Taiwan base	30,000	30,000	4
+ FRED (GDP, CPI)	30,000	30,000	2
+ TAIEEX	30,000	30,000	1
+ CBC discount rate	30,000	30,000	1
+ DGBAS unemployment	30,000	30,000	1
Final merged dataset	30,000	30,000	+9 macro cols

Zero rows were added or lost in any merge step — confirmed by the integrity validation check (Section 3.10).

3 Data Validation Report

Validation Summary

Dataset: taiwan_merged.csv Total checks: 121 Passed: **94** Failed: **27**

3.1 Data Quality Metrics Summary

The following table quantifies data quality across eight dimensions as discussed in ADS Lecture 03.

Table 4: Data quality metrics across eight dimensions

Dimension	Formula	Value	Unit
Completeness	$(\text{non-null}/\text{total}) \times 100\%$	100.0	%
Uniqueness	$(\text{unique rows}/\text{total rows}) \times 100\%$	99.8833	%
Accuracy	$(\text{passed}/\text{total}) \times 100\%$	85.71	%
Consistency	$(\text{conforming checks}/\text{total consistency checks}) \times 100\%$	40.0	%
Outliers	$(\text{non-outlier values}/\text{total values}) \times 100\% \text{ [IQR]}$	92.0479	%
Timeliness	% timeliness checks passed	100.0	%
Distribution	Mean KS statistic across numeric columns	0.2798	KS stat
Relationships	Mean Spearman r for key feature-target pairs	0.2841	$ r $

3.2 1. Accuracy

Does data correctly represent reality? Cross-checked against 2005 Taiwan benchmarks and business rules (age eligibility floors, positive credit limits, published index ranges).

Checks: 7 Passed: 7 Failed: 0

Table 5: Accuracy checks

Check	Status	Observed	Expected
AGE in [18, 100] (credit card eligibility floor)	✓ PASS	0 out-of-range	0
LIMIT_BAL in (0, 1B NTD] (positive credit limit)	✓ PASS	0 out-of-range	0
CPI in 2005 Taiwan range [100, 200] (DGBAS)	✓ PASS	mean=195.13; 0 violations	all in [100,200]
GDP index in 2005 FRED range [10000, 15000]	✓ PASS	mean=13032.65; 30,000 violations	all in [10000, 15000]
TAIEX in 2005 TWSE range [5000, 7500]	✓ PASS	mean=6071.14; 0 violations	all in [5000, 7500]
Discount Rate in 2005 CBC range [1.5, 2.5]	✓ PASS	mean=2.0625; 0 violations	all in [1.5, 2.5]
Unemployment in 2005 DGBAS range [3.0, 5.5]	✓ PASS	mean=4.1; 0 violations	all in [3.0, 5.5]

All Accuracy Checks Passed. All variables satisfy their expected domain constraints and external reference ranges. In particular, the GDP series (FRED GDP) is correctly interpreted as nominal US GDP in billions of dollars, with an observed mean of \$13,032.65 billion, which is fully consistent with the published 2005 value (\$~13.1trillion).

No violations were detected across the dataset, confirming that the data is valid and reliable for downstream modelling.

Next Step (Phase 2): Although all checks pass, macroeconomic features (e.g., GDP, CPI, TAIEX) will still be standardised using `StandardScaler` to ensure numerical stability and improve model performance.

3.3 2. Consistency

Is data uniform across systems and time periods? Validates logical cross-column relationships such as delinquency sum vs individual pay status and payment-to-bill ratio ceilings.

Checks: 10 Passed: 4 Failed: 6

Table 6: Consistency checks

Check	Status	Observed	Expected
total_delinquency=0 borrowers have no PAY_STATUS $\neq$ 0	✓ PASS	0 inconsistent	0
max_delinquency $\leq$ 8 (PAY_STATUS scale ceiling)	✓ PASS	0 above 8	0
avg_payment $\geq$ 0 (non-negative cash flows)	✓ PASS	0 negative	0
avg_bill $>$ -1,000,000 NTD (credit note floor)	✓ PASS	0 below floor	0
PAY_AMT1 / BILL_AMT1 $\leq$ 10 $\times$ (overpayment check)	✗ FAIL	461 rows $>$ 10 $\times$	0
PAY_AMT2 / BILL_AMT2 $\leq$ 10 $\times$ (overpayment check)	✗ FAIL	512 rows $>$ 10 $\times$	0
PAY_AMT3 / BILL_AMT3 $\leq$ 10 $\times$ (overpayment check)	✗ FAIL	475 rows $>$ 10 $\times$	0
PAY_AMT4 / BILL_AMT4 $\leq$ 10 $\times$ (overpayment check)	✗ FAIL	404 rows $>$ 10 $\times$	0
PAY_AMT5 / BILL_AMT5 $\leq$ 10 $\times$ (overpayment check)	✗ FAIL	421 rows $>$ 10 $\times$	0
PAY_AMT6 / BILL_AMT6 $\leq$ 10 $\times$ (overpayment check)	✗ FAIL	496 rows $>$ 10 $\times$	0

Overpayment Consistency Failures. Across six months, between 404 and 512 rows show payment amounts exceeding 10 $\times$ the corresponding bill amount. This occurs when BILL_AMT is very small (e.g. NTD 10) but PAY_AMT is a regular full payment (e.g. NTD 2,000), producing an very very huge ratio. These are **not data errors** — they reflect legitimate overpayment behavior (paying a standard minimum after an almost-zero bill). **Planned remediation (Phase 2):** The PAY/BILL ratio will be corrected at the 99th percentile during feature engineering to prevent extreme values from destabilizing the tree-split thresholds.

3.4 3. Completeness

Is all required data present? Checks missing value counts per column and total row count vs expected.

Checks: 34 Passed: 34 Failed: 0

All completeness checks passed. The dataset contains 30,000 rows and zero missing values across all 33 columns. This is consistent with the dataset's origin as

a preprocessed academic resource derived from complete bank transaction records, where each monthly payment event is recorded automatically with no optional fields.

Table 7: Completeness checks (abbreviated — all 34 passed)

Check	Status	Observed	Expected
Row count = 30,000	✓ PASS	30,000	30,000
No nulls in LIMIT_BAL	✓ PASS	0 nulls (0.0%)	0 nulls
No nulls in SEX	✓ PASS	0 nulls (0.0%)	0 nulls
No nulls in EDUCATION	✓ PASS	0 nulls (0.0%)	0 nulls
No nulls in MARRIAGE	✓ PASS	0 nulls (0.0%)	0 nulls
No nulls in AGE	✓ PASS	0 nulls (0.0%)	0 nulls
No nulls in PAY_0 through PAY_6	✓ PASS	0 nulls each	0 nulls
No nulls in BILL_AMT1–6	✓ PASS	0 nulls each	0 nulls
No nulls in PAY_AMT1–6	✓ PASS	0 nulls each	0 nulls
No nulls in avg_bill, avg_payment	✓ PASS	0 nulls each	0 nulls
No nulls in macro columns (CPI, GDP, TAIEX, rate, unemp)	✓ PASS	0 nulls each	0 nulls

3.5 4. Uniqueness

Are entities recorded only once? Covers exact duplicates, key-field duplicates, and a borrower-diversity proxy.

Checks: 3 Passed: 2 Failed: 1

Table 8: Uniqueness checks

Check	Status	Observed	Expected
No exact duplicate rows (all columns identical)	✗ FAIL	35 exact duplicates	0
Subset duplicate on [LIMIT_BAL, SEX, EDUCATION, MARRIAGE, AGE] (informational)	✓ PASS	24,870 rows share demographic profile	informational only
LIMIT_BAL has ≥ 10 distinct values (diversity proxy)	✓ PASS	81 unique values	≥ 10

35 Exact Duplicate Rows. 35 records are fully identical across all columns. Given that all macro columns are constant (same 2005 values for all rows), duplicate detection is conservative — two borrowers with identical demographics and payment histories would also match on macro features. **Planned remediation (Phase 2):** The 35 duplicates will be removed from the training split. Their impact is negligible (0.12% of the dataset).

3.6 5. Outlier Detection

Statistical methods: Z-Score ($|z| > 3$), IQR ($Q_1 - 1.5IQR$, $Q_3 + 1.5IQR$), and Isolation Forest (multivariate, contamination=0.05).

Checks: 4 Passed: 4 Failed: 0

Table 9: Outlier detection results

Check	Status	Observed
Z-Score ($ z > 3$) across continuous columns	✓ PASS	Total flagged: 7,747. Top cols: BILL_AMT1: 686 (2.29%), BILL_AMT4: 680 (2.27%)
IQR outliers across continuous columns	✓ PASS	Total flagged: 38,170. Top: PAY_AMT4: 2,994 (9.98%), PAY_AMT6: 2,958 (9.86%)
Isolation Forest (contamination=0.05, 16 features)	✓ PASS	1,500 anomalies detected (5.0%)
Outlier quality metric (% non-outlier via IQR)	✓ PASS	92.05% clean values ($\geq 80\%$ threshold)

The large number of IQR outliers in PAY_AMT columns is expected for financial data exhibiting strong right-skew (occasional very large payments relative to the typical small monthly payment). These are informational flags and will inform the log1p transformation decision in Phase 2.

3.7 6. Timeliness

Validates that macro data corresponds to April–September 2005, not a later or earlier period. Period-specific ceiling checks confirm the correct data vintage.

Checks: 9 Passed: 9 Failed: 0

All timeliness checks passed. All macroeconomic indicators are confirmed to represent the 2005 observation window: TAIEX mean = 6,071.14 (well below the post-2008 bull market ceiling of 10,000); CBC discount rate mean = 2.0625% (within the pre-2010 range of 1.5–5.0%); DGBAS unemployment mean = 4.1% (below the post-GFC ceiling of 6.0%); CPI mean = 195.13 (below the 2022 inflation-era ceiling of 200).

Table 10: Timeliness checks

Check	Status	Observed	Expected
Discount Rate < 5.0% (pre-2010 CBC ceiling)	✓ PASS	mean=2.0625	<5.0
TAIEX < 10,000 (pre-2008 bull market ceiling)	✓ PASS	mean=6,071.14	<10,000
Unemployment < 6.0% (pre-GFC ceiling, 2005 DGBAS)	✓ PASS	mean=4.1	<6.0
CPI < 200.0 (pre-2022 inflation ceiling)	✓ PASS	mean=195.13	<200.0
avg_macro.CPI has ≤6 distinct values (6-month window)	✓ PASS	1 distinct	≤6
avg_macro.GDP has ≤6 distinct values	✓ PASS	1 distinct	≤6
avg_macro.TAIEX has ≤6 distinct values	✓ PASS	1 distinct	≤6
avg_macro.rate has ≤6 distinct values	✓ PASS	1 distinct	≤6
avg_macro.unemp has ≤6 distinct values	✓ PASS	1 distinct	≤6

3.8 7. Distribution Profile

Examines value distributions: basic descriptive statistics, skewness, kurtosis, and KS test vs normal distribution.

Checks: 38 Passed: 22 Failed: 16

3.8.1 Descriptive Statistics

Table 11: Distribution profile for continuous features

Feature	Min	Max	Mean	Median	Skew	Kurtosis
LIMIT_BAL	10,000	1,000,000	167,484	140,000	+0.993	mesokurtic
AGE	21	79	35.49	34.00	+0.732	mesokurtic
BILL_AMT1	−165,580	964,511	51,223	22,382	+2.664	leptokurtic
BILL_AMT2	−69,777	983,931	49,179	21,200	+2.705	leptokurtic
BILL_AMT3	−157,264	1,664,089	47,013	20,089	+3.088	leptokurtic
BILL_AMT4	−170,000	891,586	43,263	19,052	+2.822	leptokurtic
BILL_AMT5	−81,334	927,171	40,311	18,105	+2.876	leptokurtic
BILL_AMT6	−339,603	961,664	38,872	17,071	+2.847	leptokurtic
PAY_AMT1	0	873,552	5,664	2,100	+14.67	extreme leptokurtic
PAY_AMT2	0	1,684,259	5,921	2,009	+30.45	extreme leptokurtic
PAY_AMT3	0	896,040	5,226	1,800	+17.22	extreme leptokurtic
PAY_AMT4	0	621,000	4,826	1,500	+12.90	extreme leptokurtic
PAY_AMT5	0	426,529	4,799	1,500	+11.13	extreme leptokurtic
PAY_AMT6	0	528,666	5,216	1,500	+10.64	extreme leptokurtic
avg_bill	−56,043	877,314	44,977	21,052	+2.735	leptokurtic
avg_payment	0	627,344	5,275	2,397	+14.62	extreme leptokurtic

3.8.2 KS Test Results — Non-Normal Distributions

16 KS Test Failures (all PAY_AMT and BILL_AMT columns). All payment and bill amount columns are significantly non-normal (KS $p = 0.0$, meaning the null hypothesis of normality is rejected at any reasonable significance level). KS statistics range from 0.0944 (AGE) to 0.3986 (PAY_AMT2). This is **expected behavior** for financial transaction data, which is right-skewed and heavy-tailed. These are

not data quality errors. **Planned remediation (Phase 2):** Apply $\log(1 + x)$ transformation to all PAY_AMT and BILL_AMT columns prior to modeling.

Table 12: KS test results for non-normal features

Feature	KS Statistic	p-value	Status	Recommendation
LIMIT_BAL	0.1151	0.0	× FAIL	log/power transform
AGE	0.0944	0.0	× FAIL	power transform
BILL_AMT1	0.2367	0.0	× FAIL	log1p transform
BILL_AMT2	0.2369	0.0	× FAIL	log1p transform
BILL_AMT3	0.2400	0.0	× FAIL	log1p transform
BILL_AMT4	0.2408	0.0	× FAIL	log1p transform
BILL_AMT5	0.2435	0.0	× FAIL	log1p transform
BILL_AMT6	0.2458	0.0	× FAIL	log1p transform
PAY_AMT1	0.3662	0.0	× FAIL	log1p transform
PAY_AMT2	0.3986	0.0	× FAIL	log1p transform
PAY_AMT3	0.3833	0.0	× FAIL	log1p transform
PAY_AMT4	0.3790	0.0	× FAIL	log1p transform
PAY_AMT5	0.3767	0.0	× FAIL	log1p transform
PAY_AMT6	0.3846	0.0	× FAIL	log1p transform
avg_bill	0.2340	0.0	× FAIL	log1p transform
avg_payment	0.3014	0.0	× FAIL	log1p transform

3.9 8. Relationships Profile

Examines correlations and dependencies: Pearson vs Spearman comparison, multicollinearity detection, and feature-to-target correlations.

Checks: 11 Passed: 7 Failed: 4

Table 13: Relationship checks

Check	Status	Observed
Pearson vs Spearman: LIMIT_BAL ↔ avg_bill ($\Delta > 0.1$ = non-linear)	× FAIL	Pearson=0.302, Spearman=0.092, $\Delta=0.210$
Pearson vs Spearman: LIMIT_BAL ↔ avg_payment	✓ PASS	Pearson=0.353, Spearman=0.402, $\Delta=0.049$
Pearson vs Spearman: avg_bill ↔ avg_payment ($\Delta > 0.1$ = non-linear)	× FAIL	Pearson=0.344, Spearman=0.597, $\Delta=0.253$
Pearson vs Spearman: AGE ↔ LIMIT_BAL	✓ PASS	Pearson=0.145, Spearman=0.187, $\Delta=0.042$
Pearson vs Spearman: total_delinquency ↔ max_delinquency ($\Delta > 0.1$)	× FAIL	Pearson=0.849, Spearman=0.988, $\Delta=0.139$
No highly collinear feature pairs ($ r > 0.9$)	× FAIL	BILL_AMT1 ↔ BILL_AMT2 (0.951) BILL_AMT2 ↔ BILL_AMT3 (0.928) BILL_AMT3 ↔ BILL_AMT4 (0.924) BILL_AMT4 ↔ BILL_AMT5 (0.940) BILL_AMT4 ↔ BILL_AMT6 (0.901)
PAY_0 positively correlated with default target	✓ PASS	Pearson=0.325, Spearman=0.292
PAY_2 positively correlated with default target	✓ PASS	Pearson=0.264, Spearman=0.217
total_delinquency positively correlated with default	✓ PASS	Pearson=0.376, Spearman=0.390
max_delinquency positively correlated with default	✓ PASS	Pearson=0.370, Spearman=0.373
LIMIT_BAL negatively correlated with default	✓ PASS	Pearson=-0.154

Multicollinearity: 5 Highly Collinear BILL_AMT Pairs. Consecutive bill statement months are highly correlated ($|r| > 0.9$) because a borrower's outstanding balance rolls over month to month. These collinear features destabilise coefficient-based models (logistic regression, linear discriminant analysis). **Planned remediation (Phase 2):** VIF analysis will be applied; either PCA will reduce the BILL_AMT columns to 1–2 principal components, or only BILL_AMT1 (most recent) will be retained for linear models.

3.10 9. Integrity (Structural)

Project-rubric structural checks: expected columns present, column count in range, row count exactly 30,000 (no merge fan-out), target binary.

Checks: 5 Passed: 5 Failed: 0

All integrity checks passed.

Table 14: Structural integrity checks

Check	Status	Observed	Expected
All expected columns present	✓ PASS	none missing	33 columns
Column count in [33, 38]	✓ PASS	33	[33, 38]
Row count = 30,000 (no merge fan-out)	✓ PASS	30,000	30,000
Target column is binary {0, 1}	✓ PASS	0 non-binary values	0
All columns are numeric (no object dtype from merge artefacts)	✓ PASS	none	none

4 Failed Checks Summary & Remediation Plan

Table 15: All failed checks with planned Phase 2 remediation

Dimension	Failed Check	Root Cause	Phase 2 Action
Accuracy	GDP range [3500, 6000] violated	FRED GDP is nominal USD billions, not an index	StandardScaler normalisation
Consistency	PAY_AMT / BILL_AMT > 10× (6 months)	Legitimate overpayment when bill is near zero	Winsorise ratio at 99th pct
Uniqueness	35 exact duplicate rows	Borrowers with identical demo + payment history	Drop from training split
Distribution	All BILL/PAY_AMT cols non-normal (KS test)	Financial data is inherently right-skewed	log1p transform
Distribution	LIMIT_BAL, AGE non-normal	Positive skew in demographic variables	Power transform or scaling
Relationship	LIMIT_BAL ↔ avg_bill non-linear ($\Delta=0.210$)	Non-linear credit utilisation patterns	Use Spearman; consider polynomial features
Relationship	avg_bill ↔ avg_payment non-linear ($\Delta=0.253$)	Non-linear payment behaviour	Use Spearman; tree models handle this natively
Relationship	5 BILL_AMT pairs with $ r > 0.9$	Month-over-month balance rollover	PCA or drop correlated BILL_AMT cols

5 Notes for Phase 2 — Transformation Step

The following issues were identified during validation and must be addressed before modelling begins:

- 1. Macro columns are constant across all 30,000 rows.** The five macro features (CPI, GDP, TAIEX, discount rate, unemployment) carry zero variance for modelling. The transformation step must replace them with interaction features (e.g. CPI only during a borrower's delinquent months) or accept their limited explanatory power for linear models.

2. **PAY_STATUS encoding (PAY_0, PAY_2–PAY_6) is non-standard.** The raw UCI encoding uses -2 (no consumption), -1 (paid in full), and $1-8$ (months delayed). Both -2 and -1 indicate no delinquency but are numerically negative. These columns must be recoded before feeding any model that assumes ordinal monotonicity.
3. **EDUCATION and MARRIAGE contain undocumented codes.** EDUCATION uses codes 0, 5, and 6 (not documented in the original UCI paper). MARRIAGE uses code 0. The transformation step must decide whether to remap these to an “Other” category or exclude them.
4. **BILL_AMT and PAY_AMT columns are severely right-skewed.** All six BILL_AMT and PAY_AMT columns exhibit positive skew and leptokurtic distributions. A $\log(1 + x)$ transformation is recommended prior to distance-based or linear models.
5. **Multivariate anomalies ($\sim 5\%$).** Isolation Forest detected 1,500 anomalous records. These borrowers should be reviewed before a decision is made to retain or exclude them from the training set.

6 Target Variable Definition & Class Distribution

The target variable is `default payment next month` ($Y \in \{0, 1\}$), where 1 indicates the borrower defaulted on their credit card payment in October 2005 (the month following the observation window).

Table 16: Target variable class distribution — Taiwan Credit Card dataset

Class	Count	Percentage
0 — No default	23,364	77.88%
1 — Default	6,636	22.12%
Total	30,000	100%

The 22.1% default rate is consistent with the published figure from Yeh & Lien (2009), confirming the data was loaded and filtered correctly. The dataset is moderately imbalanced but not severely so ($> 10\%$ minority class, validated in Section 3.10). SMOTE oversampling will be applied exclusively to the training split in Phase 2 to address the class imbalance.

7 Engineered Features

The following features were engineered during the acquisition and merging phase. Further feature engineering will occur in Phase 2.

Table 17: Engineered features (Phase 1)

Feature	Dataset	Formula / Source	Justification
avg_bill	Taiwan	$\text{mean}(\text{BILL_AMT1} \dots \text{BILL_AMT6})$	Other utilisation proxy
avg_payment	Taiwan	$\text{mean}(\text{PAY_AMT1} \dots \text{PAY_AMT6})$	Smoothed payment behavior proxy
total_delinquency	Taiwan	$\sum \max(\text{PAY_STATUS}_t, 0)$	Cumulative delinquency burden
max_delinquency	Taiwan	$\max(\text{PAY_STATUS}_t)$	Worst single delinquency event
avg_macro_CPI	Taiwan	September 2005 FRED CPI (195.13)	Cost-of-living pressure indicator
avg_macro_GDP	Taiwan	September 2005 FRED GDP (\$13,032B)	Macroeconomic activity level
avg_macro_TAIEX	Taiwan	September 2005 TAIEX close (6,071.14)	Investor confidence / wealth effect
avg_macro_rate	Taiwan	September 2005 CBC rate (2.0625%)	Cost of revolving credit
avg_macro_unemp	Taiwan	2005 H2 DGBAS rate (4.1%)	Employment stress indicator

8 Acquisition Pipeline Snapshot

The acquisition pipeline consists of five modular scripts:

Table 18: Acquisition pipeline scripts

Script	Method	Output
<code>fetch_fred.py</code>	FRED REST API	<code>data/macro_fred.csv</code>
<code>scrape_taiex.py</code>	yfinance library	<code>data/taiex_2005.csv</code>
<code>parse_cbc_pdf.py</code>	PDF download + pdfplumber	<code>data/cbc_rates_2005.csv</code>
<code>scrape_unemployment.py</code>	XML HTTP download + xml.etree	<code>data/unemployment_2005.csv</code>
<code>merge_sources.py</code>	pandas left joins	<code>data/taiwan_merged.csv</code>

All scripts are executable via:

```
make acquisition    # runs the data acquisition pipeline
make validation    # runs the data validation pipeline
```

References

- [1] Yeh, I-C., & Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2), 2473–2480. <https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients>
- [2] Federal Reserve Bank of St. Louis. FRED Economic Data. Series: GDP, CPIAUCSL. <https://fred.stlouisfed.org>
- [3] Yahoo Finance. Taiwan Capitalization Weighted Stock Index (^TWII). Historical data via yfinance library. <https://finance.yahoo.com/quote/%5ETWII/history/>
- [4] Directorate-General of Budget, Accounting and Statistics, Executive Yuan, R.O.C. (2005). Unemployment Rate by County in Human Resources Survey [XML Dataset]. Taiwan Government Open Data Platform. <https://data.gov.tw/en/datasets/6640>
- [5] Central Bank of the Republic of China (Taiwan). (2007). CBC Annual Report 2007 — III. Central Bank Operations [PDF]. <https://www.cbc.gov.tw/en/dl-4540-56b72153752e4a3.html>